

The Invented Inventor: Adapting Patent Law to Generative AI

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As artificial intelligence increasingly drives processes of discovery, the concept of *inventor* – a keystone of the modern patent system – faces severe strain. Recent judicial decisions, such as the Swiss Federal Administrative Court’s 2025 DABUS ruling, expose a deepening conceptual tension: courts formally demand intellectual creation by a natural person, even as human contributions to AI-assisted discovery become increasingly nominal. By applying the method of pragmatic genealogy, this paper reconstructs the historical “need matrices” that forged the concept of inventorship, tracing its evolution from Venetian guild economics through Romantic genius ideology to corporate R&D. This genealogy reveals that the concept of inventor is an overburdened bundle of four inferential roles – attributing ownership, desert, patent entitlement, and the responsibility to disclose – which in turn serve four distinct social functions: incentivising innovation, legitimating monopolies, disseminating knowledge, and resolving priority disputes. Ultimately, the genealogy exposes a fundamental mismatch between our inherited concept of inventor and the emerging realities of AI-driven discovery. To resolve this, I argue that we must disaggregate the concept of inventorship and develop specialised conceptual resources for each of these functions. If we invented the notion of inventor to perform certain functions, we can reinvent it to perform them better.

Keywords: intellectual property (IP); artificial intelligence (AI); patent law; inventorship; pragmatic genealogy; need matrices; disaggregation; conceptual fission.

1. A Load-Bearing Concept Under Strain

We are witnessing a structural transformation in the production of new knowledge. AI systems are increasingly discovering novel molecules, designs, and materials with unprecedented properties.¹ The AI model GNoME alone recently discovered 2.2 million materials, expanding humanity’s repository of stable materials by an order of magnitude (Merchant et al. 2023). This paradigm shift forces patent systems worldwide to confront an unprecedented question: can AI qualify as an inventor? This is rapidly emerging as one of the most consequential issues in

¹ A 2023 review in *Nature* already found AI’s impact on scientific discovery to be substantial (Wang et al. 2023), and it has accelerated rapidly since – as exemplified by the leap from the original AlphaFold to the *Isomorphic Labs Drug Design Engine* (Isomorphic Labs Team 2026). This transformation notably spans pharmaceutical R&D (Doron et al. 2025), materials science (Madika et al. 2025), and the rise of autonomous, “self-driving” laboratories that automate research across chemistry, biology, and materials (Tobias and Wahab 2025).

contemporary legal and political theory. The concept of inventorship serves as the load-bearing pillar in a system that channels trillions of dollars in economic activity, dictates the pace of medical and technological progress, and determines the distribution of the fruits of innovation.

Already, that system is buckling under severe strain.² Since 2019, Stephen Thaler has spearheaded a global campaign to have his AI system DABUS – Device for the Autonomous Bootstrapping of Unified Sentience – recognised as the sole inventor of a fractal-geometry food container. The patent applications have invariably been denied: the European Patent Office (2020), the United Kingdom Supreme Court (2023), the United States Federal Circuit (2022), Australia’s Full Federal Court (2022), the German Federal Court of Justice (2024), and – most recently – the Swiss Federal Administrative Court (2025) have all held that only natural persons qualify as inventors (Abbott 2020). As the U.S. Federal Circuit’s August 5, 2022, opinion effectively affirmed after Thaler appealed: “the invented cannot be the inventor” (Nemec and Rann 2023).

In justifying these rejections, courts have made explicit the cognitive presuppositions built into the concept. The Swiss ruling, for instance, drew on three interlocking characterizations: the Federal Patent Court’s definition of the inventor as one who “recognized the inventive concept and developed it into an instruction for technical action through creative activity”; the prevailing doctrine’s requirement of an “intellectual act of creation” (*geistiger Schöpfungsakt*); and the Federal Supreme Court’s case law requiring “intuitive-associative activity” (*intuitive-assoziative Tätigkeit*) (Swiss Federal Administrative Court 2025, §§ 4.3–4.5).

What makes the Swiss ruling particularly revealing is not its rejection of AI inventorship so much as what it accepted as sufficient for human inventorship. The court held that Thaler *himself* qualified as inventor on the grounds that he was “involved in providing data and training DABUS, received the finished solutions and recognised that this was an invention

² To date, the focus has been on the implications for the inventive step requirement – that a patentable invention must be *non-obvious* to a “person skilled in the art” (PSA – or PHOSITA in US terminology). If AI systems can rapidly generate solutions across domains, the baseline for non-obviousness may need recalibration, because AI pushes the boundary of what counts as non-obvious to a PSA or PHOSITA (Abbott 2017; Lim 2018; Reinbold 2020; Romm 2021; Heon 2022; Merritt 2023).

eligible for protection”, and thus had “sufficient overall influence on DABUS to be considered an inventor” (Swiss Federal Administrative Court 2025, § 6.3). From the AI, the court demanded an “intellectual act of creation”, but from the human, it accepted a form of involvement that sits uneasily with any robust understanding of creative intellectual activity.

The concept of inventor is thus being formally preserved at the level of doctrine while being hollowed out at the level of application. And as LLM training pivots from training against human language use to training against the world itself – using techniques such as Reinforcement Learning against Verifiable Rewards (Novikov et al. 2025) and Test-Time Training to Discover (Yuksekgonul et al. 2026) – the stream of AI-generated discoveries is set to grow exponentially. This shift from *knowledge-sharing* to *knowledge-creating* LLMs (Cunningham 2026) will only exacerbate the existing conceptual tensions.

If the notion of inventorship remains firmly tied to individual human creation, we risk a future in which the most advanced inventions are either unprotectable – stifling the investment needed to develop them – or locked away as trade secrets, blocking the diffusion of knowledge that the patent system was built to secure. If, conversely, AI-generated inventions are made patentable without rethinking the foundations, we risk granting monopolies that lack the justification the system has always demanded.

This paper takes these tensions as its point of departure, but instead of tackling the question “Can AI be an inventor?” head-on, it raises a prior, more fundamental question: *what do we need the concept of inventorship to do for us?* “Inventor” does not pick out a natural kind whose true nature we can discover through analysis alone. It denotes a *social* kind – a concept devised to solve social problems.³ Therefore, instead of asking whether this emerging technology fits our old concept, we must ask whether this old concept fits our current needs.

To answer this, we must reconstruct the practical needs that shaped the concept’s historical formation. This historical lens will help us discern what functions we need the concept to

³ As is the concept of an *invention* itself; see Pottage & Sherman (2010) for a detailed historical study to that effect.

discharge today, and how we might adapt it to AI. The method I employ is pragmatic genealogy (Queloz 2021), sharpened with the analytical apparatus of “need matrices” (Queloz 2025b, 2026).

Applied to the concept of inventor, this method reveals that the concept bundles together several distinct social functions. I argue that while these functions could hitherto be discharged by a single concept centred on individual human creation, the advent of generative AI creates a misalignment that stretches this overburdened concept to its breaking point. I make the case for *principled disaggregation* – which acknowledges the different social functions at stake and develops specialised conceptual resources for each – as a more transparent, truthful, and responsive answer to the needs underpinning the concept of inventor. We invented the inventor; we can, and probably must, reinvent it.

2. Methodology: Pragmatic Genealogy and Need Matrices

Pragmatic genealogy has its roots in Hume’s genealogy of property rights (Queloz 2021, ch. 4). Hume sought to understand the emergence of respect for property by reconstructing the fundamental predicament that would make something like the institution of property rights indispensable. He began from a “state of nature” – a “mere philosophical fiction, which never had, and never could have any reality”, but which nonetheless “deserves our attention, because nothing can more evidently shew the origin of the virtue [of respect for property]” (Hume 2000, 3.2.2.14) – it strips away the complexities of actual societies to isolate the fundamental problem that respect for property addresses. The “state of nature” is not a conjecture about prehistory; nor is it the original condition in which God places humankind according to Locke (2003, 2.6), or even a hypothetical foil serving to exhibit life under a Hobbesian state as preferable to a life that is “nasty, brutish, and short” (Hobbes 2006, 13.9); it is an *exploratory model*: it imagines a community lacking the concept in question and asks what might drive them to invent it. Its

value lies in the perspicuity with which it allows one to discover and display the connection between a concept and basic human concerns it serves.⁴

Having identified a fundamental problem, however, pragmatic genealogy does not stop at this highly idealised representation. It historicises it through *progressive de-idealisation*: tracing how specific historical forces filled in and transformed the fundamental problem. The idealised sketch and the historical genealogy are thus stages of a single narrative arc from the abstract to the concrete and from the generic to the specific.

To carry out this progressive de-idealisation with analytical precision, I draw on the notion of a *need matrix* (Queloz 2025b, 2026). A need matrix analyses the formative pressures on a concept into three kinds of components: *concerns* are the subjective pole – what a community cares about, from its economic ambitions to its values and projects; *capacities* are the mediating factors – the abilities and limitations available for the pursuit of those concerns; *circumstances* are the objective pole – the natural and social environment in which those concerns are being pursued. A *conceptual need* arises not from any one of these components in isolation but from their conjunction.

Need matrices not only crystallise which historical features were crucial to a concept's development, but also diagnose exactly where and why a concept no longer suits its present context. The key insight I shall develop is that while the *concerns* underpinning the concept of inventor remain unchanged, the relevant *capacities* and *circumstances* are now shifting dramatically, creating a mismatch between our situation and the concept we inherited.

3. A Pragmatic Genealogy of Inventorship

Consider an idealised settlement in which artisans occasionally acquire technical knowledge that the rest of the community lacks: a way of firing clay that produces harder vessels, or a

⁴ My use of a Humean state of nature to elucidate the problem that the concept of inventorship most basically solves thus differs from the way in which Merges (2011, 32–62) invokes the state of nature to justify intellectual property, which turns on an analogy between the *res communis* of the Lockean state of nature and the public domain from which new intellectual creations draw. On the contrast between Hume's and Locke's theories of property, see Waldron (2013).

method of dyeing cloth a colour that does not fade. Each artisan must then decide whether to share the technique and lose the competitive advantage, or keep it secret and preserve the advantage. From an individual perspective, secrecy is rational. But since the same is true for every artisan, a collectively detrimental pattern emerges: valuable technical knowledge remains locked in individual workshops, cannot be built upon by others, and risks being lost when an artisan dies.

This coordination problem is the fundamental problem that the concept of inventorship addresses. Introducing the concept permits the attribution of a new social role: that of the *inventor*. For what designating someone as the inventor of something fundamentally does is to strike a *bargain*: to any artisan who shares his technique, the community grants a temporary privilege; for a fixed number of years, that artisan alone may practise the technique commercially. The concept of inventorship is the linchpin of this exchange: it identifies who is entitled to the privilege in return for bearing the obligation to disclose.

What does this concept minimally require in order to perform this role? Inventors must *possess* knowledge the community lacks; they must be able to *transmit* it; and the community must be able to *verify* that the transmission has occurred. The concept does not require any more particular assumptions about how that knowledge originated.

This idealised sketch serves the same purpose as Hume's state of nature: it identifies the bare outline of the coordination problem at the root of the notion of inventorship. But it does not take us far enough. It tells us that *some* concept of inventorship is needed; it does not tell us which *particular* features that concept should have. The features now at the centre of the AI inventorship debate – such as the requirement that an inventor must have conceived the invention through an act of intellectual creation – were accrued in response to more specific historical pressures.

3.1 Inventorship as Knowledge Introduction: The Venetian Solution to Guild Secrecy (1474)

The first step in de-idealising the model is to locate the coordination problem in the earliest documented institutional response to it. Medieval guild systems organised craft knowledge

through rigid hierarchies, enforcing secrecy through oaths, physical isolation, and draconian penalties. Glassmakers on Murano were forbidden from emigrating and threatened with severe penalties, including death (Frumkin 1945, 144); Florence, Lucca, and Genoa even paid bounties for the murder of fugitive artisans (Molà 2000, 43; Kostylo 2010, 34). Yet cities needed new techniques to maintain economic competitiveness. Venice's response was the first systematic patent statute, passed by the Senate on March 19, 1474:

There are in this city, and also there come temporarily by reason of its greatness and goodness, men from different places and most clever minds, capable of devising and inventing all manner of ingenious contrivances. And should it be provided, that the works and contrivances invented by them, others having seen them could not make them and take their honor, men of such kind would exert their minds, invent and make things which would be of no small utility and benefit to our State. (Venetian Patent Statute, 1474; translated in Mandich 1948)

The concept of inventor operative here was far broader than our modern understanding. It encompassed both those who *devised* and those who merely *imported* foreign techniques. Novelty was local: an invention needed only to be “not made heretofore in our dominion”.

Genealogical lesson: In Venice, the concept of inventor served a purely economic rationale: incentivising innovation by protecting and rewarding those who were first to put a technique to work within the jurisdiction. Inventorship was indexed to place rather than to any cognitive act of origination.

Need Matrix for Venice (1474)

Concerns	Maintain economic competitiveness; attract foreign expertise; overcome guild monopolies
Capacities	Knowledge embodied in persons; administrative capacity to grant time-limited exclusivity
Circumstances	Maritime republic dependent on technical innovation; inter-city competition for skilled artisans; guild secrecy blocking diffusion
Concept needed	<i>Inventor</i> as introducer of technical knowledge

3.2 Inventorship as Instruction: The English Duty to Teach (1331–1600)

The state-of-nature model drew attention to the importance of ensuring that new knowledge is not only acquired, but *transmitted*. This aspect is particularly pronounced in the earliest records of the English patent tradition, which charged inventors with a *duty to teach*.

The earliest instance is the Letters of Protection granted to John Kempe of Flanders by Edward III in 1331, issued for the “avowed purpose of instructing the English” (*instruendi et informandi*) in fine weaving. More significant still was the 1449 grant to John of Utynam for coloured stained glass: a twenty-year monopoly issued “strictly upon the condition” that Utynam instruct the King’s subjects in glassmaking. Under Elizabeth I, this was systematised through quotas and apprenticeship ratios in patents for foreign artisans (Hulme 1896; Bracha 2016).

The inventor, in this tradition, was functionally a *teacher*. If the patentee could not teach the invention because they did not really understand it themselves, the patent was void. The patent system’s *protection* of imported knowledge was subservient to its *transmission*.

Genealogical lesson: The English tradition sharpened the dissemination function of the concept. The inventor was defined by a duty to teach.

Need Matrix for England (1331–1600)

Concerns	Acquiring and disseminating foreign technical expertise
Capacities	The Crown had the capacity to tie patents to apprenticeship
Circumstances	English craftsmanship lagging behind the Continent in many trades
Concept needed	<i>Inventor</i> as instructor

3.3 Inventorship as a Filter Against Abuse: The Statute of Monopolies (1624)

By the turn of the seventeenth century, the Crown was increasingly abusing patents – from *litterae patentae*, “open letters”, so called because they were exposed to view – to grant monopolies on common commodities (salt, playing cards, leather) to royal favourites. The courtiers holding these patents then raised prices and stifled trade, fuelling popular hostility to monopolies (MacLeod 1988, 14–16).

The Statute of Monopolies (1624) was the Parliamentary response. It outlawed all monopolies, with one exception: patents for “the sole working or making of any manner of new manufactures within this realm, to the true and first inventor”. The concept of inventor effectively functioned to distinguish legitimate patents – which brought new industries to the realm – from mere extraction schemes (Hulme 1897; Walterscheid 1997, 12–13).

Even here, however, “inventor” still carried no requirement of cognitive origination. As affirmed in the *Clothworkers of Ipswich* case (1614), the essential requirement was that the inventor “*bringeth to and for the Commonwealth a new industry*”.

Genealogical lesson: The concept of “true and first inventor” entered statutory law as a filter against corrupt monopoly grants. Only someone who brought a genuinely new industry to the commonwealth could receive a monopoly.

Need Matrix for England (1624)

Concerns	Curbing royal abuse of monopoly grants
Capacities	Parliament’s growing capacity to constrain royal power
Circumstances	Rent-seeking courtiers monopolising common commodities and stifling trade
Concept needed	<i>Inventor</i> as legitimating filter against monopoly abuse

3.4 From Embodied Know-How to Communicable Understanding (1711–1778)

Until the early eighteenth century, technical knowledge was largely embodied rather than written down. Patentees fulfilled their duty to teach by working the invention and physically demonstrating their techniques to apprentices.

Following the Glorious Revolution (1688), however, administrative and economic changes drove a shift from somatic demonstration to textual description (Pottage and Sherman 2010). Jurisdiction over patents shifted from the Privy Council, whose reach still enabled it to physically inspect workshops, to the Law Officers of the Crown, who were based in London and lacked this monitoring capacity. The solution emerged in the 1711 patent to John Nasmith for fermenting wash from molasses: the Law Officers required a written description to distinguish his improvement from the general trade of brewing. By 1734, the “patent

specification” (MacLeod 1988, 48–51) had become standard, codifying technical knowledge in a form that accelerated its dissemination in an increasingly literate society.

This administrative shift mirrored a broader economic transition from the guild system to industrial capitalism. The power of the guilds was waning; rising capitalists were not necessarily master craftsmen; and the nature of the inventions was changing: chemical processes or mechanical devices were more readily codifiable than the skill of a master weaver.

The legal ratification of this shift was the landmark ruling in *Liardet v. Johnson* (1778), holding that a specification must be “sufficiently full and detailed to enable anyone, skilled in the art to which the invention pertained, to understand and apply it without undue experimentation”. After *Liardet*, the inventor had to describe the invention clearly enough for others to reproduce it – imposing unprecedented demands on the inventor’s understanding.

Genealogical lesson: Administrative and economic shifts transformed the inventor from a practitioner into an *explainer* – someone who needed to *understand* the invention sufficiently to articulate in writing what hitherto remained implicit in practice. The connection between inventorship and understanding did not originate in reflection about creativity, but in a shift in disclosure requirements.

Need Matrix for England (1778)	
Concerns	Administrating and defining the scope of patents with precision
Capacities	Dwindling capacity to monitor physical working of inventions; growing literacy
Circumstances	Patent administration to London-based Law Officers; weakening of guilds; industrialisation
Concept needed	<i>Inventor</i> as explainer capable of written disclosure

3.5 Invention as Labour of the Mind: The Enlightenment-Romantic Transformation (1689–1870)

Between 1689 and 1870, the concept of inventor underwent its most consequential transformation – one that turned the *justification* for monopolies from an instrumental into a non-instrumental one.

Locke's labour theory of property (1689) provided the idea that individuals acquire property by mixing their labour with common resources. Whether Locke intended this to extend to intellectual property is debated (Merges 2011), but the extension was realised by American jurists: Justice Woodbury in *Davoll v. Brown* (1845) declared: "we protect intellectual property, the labors of the mind", which are "as much the fruit" of "honest industry" as "wheat". Within Locke's framework, it became possible to *make sense* of patents not simply as state-granted privileges, but as *natural property rights* (Mossoff 2012).⁵

Once the French Revolution abolished guilds and royal privileges, patents needed to be recast in terms of natural property rights if they were to be preserved (Baudry 2020). The French law of 1791 accordingly declared: "All new discoveries are the property of the author" (Hilaire-Pérez 2010). The U.S. Constitution took a more instrumental path, conceptualising patents as means "to promote the progress of science and useful arts" (Art. 1, § 8, cl. 8) – but this coexisted uneasily with the natural-property culture (Walterscheid 1997; Bracha 2016).

The link between individual creative labour and ownership of its product was then affirmed by Romanticism. As Martha Woodmansee (1984, 1992) argues, the modern concept of *author* as sole creator only emerged in the mid-eighteenth century, with the rise of individuals earning their livelihood from writing. But this Romantic reconceptualisation of creative production carried over from literary authorship to technological invention. Invention came to be seen as an expression of the creator's *personality*, as the influential Hegelian theory had it (Hegel 1970). Inventors rose to heroic stature in Victorian Britain (MacLeod 2007). During the "patent controversy" of the 1860s–70s, defenders of patents assimilated invention to literary creation: "Watt may be said to have created his particular steam-engine in the same sense that Milton may be said to have created *Paradise Lost*" (Sherman and Bently 1999, 150).

Genealogical lesson: The Enlightenment-Romantic transformation altered the *basis of legitimation* for patent monopolies. Lockean ideas licensed the inference from inventorship to

⁵ On the real political constraints imposed by the requirements on legitimation stories to *make sense* to their addressees, see Williams (2005).

ownership: intellectual labour produces a natural property right. Romanticism supplemented this with the inference from inventorship to *desert*: the inventor's genial originality *deserves* honour (*Erfinderehre*) and reward. Together, these two inferences both *narrowed* and *individualised* the notion of inventorship: if the labour of creative origination is what justifies a monopoly, then importation no longer qualifies; and if invention is an act of creative genius that essentially expresses the creator's personality, then inventorship must be attributed to a specific individual.

Need Matrix for the Enlightenment-Romantic Transformation (1689–1870)

Concerns	Preserve the patent system in republics without royal grants
Capacities	Ability to see individual intellectual labour as a basis for natural property rights
Circumstances	Abolition of guilds and royal privileges; anti-patent movement of 1860-70; influence of creative genius model from literary authorship
Concept needed	<i>Inventor</i> as creative originator deserving recognition and reward

3.6 Corporate R&D and the Doctrine of Conception (1876–1952)

The Romantic archetype of the individual genius soon clashed with industrial reality, however. Thomas Edison's "invention factory" at Menlo Park (1876) industrialised invention through multidisciplinary collaboration. Yet patents continued to be filed with him as sole inventor. By the 1930s, over a thousand corporate research laboratories had been established (Usselman 2013, 9).

This created a tension prefiguring the AI challenge: patent law required an individual inventor when inventions increasingly emerged from teams (Khan 2019). The response was what might be called *preservation through substantive thinning*: the substance of the concept was thinned out by reducing the entire complex process of invention to a single mental act – the moment of *conception*. In *Townsend v. Smith* (1929), conception was defined as "the formation in the mind of the inventor of a definite and permanent idea of the complete and operative invention".

On the one hand, this provided a way to attribute inventions to individuals even when they were the work of teams. But it also answered to a more basic need to *resolve priority disputes*. Most jurisdictions have a “first-to-file” system: whoever *files* first wins. But the U.S., in accordance with the non-instrumental legitimation for monopolies, adopted a “first-to-invent” system: whoever *invented* first wins. Courts therefore needed a datable mental event that could be attributed to a specific individual – and the conception doctrine supplied it.

The high-water mark of this strategy was the “flash of genius” test articulated in *Cuno Engineering Corp. v. Automatic Devices Corp.* (1941), when Justice Douglas held that a new device “must reveal the flash of creative genius, not merely the skill of the calling”. This laid bare the conception doctrine’s Romantic roots. Succumbing to pressure from the decidedly less Romantic reality of corporate R&D, however, the “flash of genius” criterion was overturned in 1952, replaced by the criterion of *non-obviousness* operative today.

Genealogical lesson: Just as inventive practice was being industrialised, the individualisation of inventorship was crystallised in legal doctrine with the doctrine of conception, which answered to the need to resolve priority disputes under a “first-to-invent” system.

Need Matrix for the United States (1929)

Concerns	Ensuring patentability of corporate inventions; resolving priority disputes
Capacities	Research teams producing inventions no single individual could create alone
Circumstances	Rise of corporate R&D; first-to-invent system requiring temporal ordering
Concept needed	Inventorship as “conception” – a particular individual’s datable cognitive act

4. Diagnosing Inferential Roles and Social Functions

This genealogy tells us what to look for in the concept of inventor today. To do this with precision, however, we need a distinction that the genealogy is well placed to illuminate: the distinction between a concept’s *inferential roles* and its *social functions*. A concept’s *inferential role* is what it immediately allows concept-users to infer or do – its manifest content in reasoning. A concept’s *social function* is what it tends to achieve *by* playing this inferential role

– systematic practical consequences that concept-users need not be aware of. The coming apart of inferential roles from social functions is what makes it possible for a concept to appear to be working well at the inferential level while its social functions are being systematically undermined – which, I shall argue, is precisely what is happening to the concept of inventor.

4.1 The Four Inferential Roles

The concept of inventorship most basically attributes an invention *to someone*. In doing so, it licenses four inferences. From “X is the inventor of Y”, one can infer that:

- (i) X *owns* Y – the invention is X’s intellectual property, a product of X’s mind in an analogous sense to that in which physical objects produced by one’s labour are one’s property.
- (ii) X *deserves* recognition and reward for the achievement – the creative intellectual labour invested in the invention generates a moral claim to honour (*Erfinderehre*) and compensation.
- (iii) X is *entitled to the patent* – X holds the right to exclude others from practising the invention for the duration of the patent term.
- (iv) X bears a *responsibility to disclose* – the obligation to describe the invention in sufficient detail that others skilled in the art can reproduce and build upon it, which in turn requires understanding the invention sufficiently to do this in writing.

In the context in which the concept was constructed, all four inferences pointed to the same individual. The person who originated the invention was typically the person who owned it, who deserved recognition for it, who held the patent, and who could explain how it worked.

4.2 The Four Social Functions

As the genealogy reveals, however, the concept bundling these inferences also serves less obvious social functions – functions that the concept discharges *by* licensing these inferences.

- (1) *The incentivisation function.* By attributing ownership, desert, and entitlement to specific agents, the concept incentivises the introduction of novel technical knowledge – the

concern that has animated patent systems since the Venetian statute's effort to attract "clever minds" to the republic.

- (2) *The legitimization function.* By attributing ownership as natural property and desert as a moral claim, the concept legitimates what would otherwise be a suspect restraint on competition, justifying exceptions to the default rule that granting monopolies is illegal. This function became explicit with the Statute of Monopolies and was deepened by the Enlightenment-Romantic transformation, which supplemented the originally instrumental justification with a non-instrumental one grounded in creative intellectual labour.
- (3) *The dissemination function.* By attributing responsibility to disclose, the concept helps to ensure that technical knowledge gets diffused as widely as possible, so that others can reproduce and build upon it. Of course, the availability of the concept is not by itself a guarantee: Antonie van Leeuwenhoek (1632–1723), despite living in one of the first nations to develop a patent system, took the secret of his single-lens microscopes to his grave, thereby holding back microbiology for over 150 years – which is how long it took for his lenses to be matched (Cocquyt et al. 2021). But in aggregate, the concept promoted the dissemination of knowledge by making its protection conditional on disclosure. This is the function most directly continuous with the fundamental coordination problem identified in the state-of-nature model.
- (4) *The dispute resolution function.* By tying inventorship to a cognitive act attributable to a specific individual at a specific time, the concept provides a mechanism for resolving priority disputes between claimants to a monopoly. Under the U.S. first-to-invent system, the datable mental event of "conception" supplied the temporal ordering criterion.

The genealogy explains why the concept accumulated these particular inferential roles and social functions. But it also reveals that several of the historical reasons for the concept's current features have now lapsed. Technical knowledge is no longer primarily embodied in persons. Guilds and royal prerogatives have all but disappeared. The first-to-invent system, which drove

the narrowing of inventorship to a datable cognitive act, was abandoned in 2013. And the Romantic genius model is precisely what the advent of creative AI models puts under pressure. Distinguishing which features were accrued for reasons that remain reasons for us today, and which for reasons that have lapsed, is the work of the next section.

5. How AI Disrupts the Four Functions

As AI systems increasingly become capable of making genuine discoveries, we are shifting from a world of invention through individual creation to one of invention through curation and direction. Unlike earlier, domain-specific systems such as AlphaFold, the latest generation of AI discovery techniques – AlphaEvolve, Aletheia, TTT-Discover – employ general methods that can be rapidly adapted to new domains (Cunningham 2026). And while existing IP frameworks accommodate AI *assistance* comfortably, they struggle to accommodate the increasingly common scenario in which the inventive leap originates in AI processing rather than in human intellectual activity.

If inventorship remains firmly tied to individual human creation, the concept risks ceasing to perform some or all of its animating functions. This may not be obvious, since the concept may still appear to work well in practice: a February 2025 report from the Swiss government, for example, concluded that “there is currently no need for regulation in patent law”, since “the number of patent applications for AI-based inventions has been increasing exponentially and does not appear to present any challenges”; the report takes this to indicate “that the system is functioning satisfactorily” (§ 4.3.3). This is a remarkable instance of a system measuring its own health by its throughput rather than by the quality of its conceptual operations.

The distinction between inferential roles and social functions explains why the threat to the concept’s proper working is not necessarily manifest. By stretching the concept, courts can continue to grant patents to individual human inventors.

But for the concept to fail us is not necessarily for it to become inapplicable or for the patent system to grind to a halt. The problem can be the less obvious one – rendered salient only by a pragmatic genealogy – that the concept of inventor has ceased to perform its social functions.

This will be so if its *inferential* consequences no longer reliably have the desired *causal* consequences: if ownership-attribution no longer reliably translates to dispute resolution, desert-attribution to legitimation, entitlement-conferral to incentivisation, and disclosure-obligation to dissemination. What AI threatens to disrupt, therefore, is the reliability of the inference *from* the concept's performance of its inferential roles *to* the performance of its social functions. Let me now show this for each of the four functions in turn.

5.1 Incentivisation

The incentivisation function is threatened whether or not AI-generated inventions are treated as patentable – but in different ways.⁶

If AI-generated inventions are *not* patentable – the position supported by the inherited concept, and which courts around the world have converged on – the incentive mechanism is simply removed. Firms will be unable to recoup their investments through temporary exclusivity. Yet even AI-discovered materials must still be prototyped and tested; AI-discovered drug candidates must still pass prohibitively expensive clinical trials.

If, on the other hand, AI-generated inventions *are* patentable, the incentive mechanism is preserved, but misdirected. The inherited concept incentivises the act of *conception* – the moment at which an inventive idea crystallises in an individual mind. But once knowledge-creating AI systems are in principle deployable at scale, the bottleneck to innovation is no longer conception itself; the bottleneck shifts *upstream* of conception: to the compute infrastructure required to train and run discovery systems, the expert curation of high-quality datasets, the design of verification procedures that can distinguish genuine breakthroughs from noise, and the domain expertise needed to identify which problems are worth tackling. Meanwhile, the bottlenecks *downstream* of conception, such as clinical trials and other forms of testing, remain. These upstream and downstream investments are the activities that need incentivising, yet the

⁶ Just how effective patent-based incentivisation is in practice is the subject of some debate – see Hettinger (1989) and Moore (2002) for theoretical challenges, and see Moser (2013), Boldrin & Levine (2013), and Baudry & Dumont (2017) for reviews of the empirical evidence.

inherited concept routes incentivisation through a bottleneck in the process of innovation that no longer corresponds to where the real hurdles now lie.

The economics of AI-generated discovery compound the problem. General-purpose AI discovery methods can be rapidly deployed across domains, making the marginal cost of any *particular* discovery low relative to the high fixed costs of building the discovery infrastructure. Cunningham (2026) identifies the likely consequence: an “IP land-grab” in which firms race to patent thousands of AI-generated discoveries at negligible marginal cost. Granting a twenty-year monopoly on each of those discoveries risks significantly over-incentivising individual inventions. What each firm needs – and what the patent system currently fails to provide – is the promise of a return calibrated to the *infrastructure* investment rather than to the marginal costs of individual discoveries.

5.2 Dissemination

The dissemination function, too, is threatened on both policy paths, though in different ways. If AI-generated inventions are *not* patentable, companies have strong incentives to lock them up as trade secrets. The patent bargain exists to draw technical knowledge out of secrecy and into the public domain. Without it, the most advanced inventions that could benefit entire industries are more likely to remain proprietary indefinitely.

If AI-generated inventions *are* patentable, the dissemination function faces a subtler but no less serious threat. As the genealogy showed, the shift from requiring that the invention be *worked* to requiring that it be *taught* to requiring that it be *specified in writing* forged a link between inventorship and understanding. Understanding was demanded not for its own sake, but because it was a precondition for producing adequate written specifications.

With AI systems, however, this transmission of understanding from the inventor to the wider public is set to be more difficult. Even where AI systems develop internal organisations that sustain comparison with human understanding (Beckmann and Queloz 2025), the *forms* of understanding they possess tend towards opacity – high predictive power without the principled simplicity that characterises transmissible understanding (Queloz and Beckmann

2026). The processes through which AI systems learn lack the social pressures that press human understanding into a communicable form. The disclosure that AI “inventors” can provide may therefore fall short of conveying the principled and actionable technical understanding that the patent system aims to disseminate.

Moreover, when the knowledge-creating process is *itself* an innovative general-purpose technology, the dissemination challenge extends beyond disclosing the discovery to ensuring that the technology enabling the discovery is itself sufficiently disclosed to prevent the monopolisation of general-purpose research tools. Yet, as Cunningham (2026) notes, current incentives are such that AI firms may prefer to capitalise on the individual discoveries made by their general-purpose discovery technology, or even offer a service providing individual innovations on demand, without exposing the underlying discovery technology itself. Our inherited framework, focused on disseminating the individual inventor’s understanding of a particular invention, is ill-equipped to address this unprecedented challenge.

5.3 Legitimation

The legitimation function is threatened if AI-generated inventions are treated as patentable. Under the inherited concept of inventor, a monopoly is justified by the creative intellectual labour of the inventor. But, as the Swiss DABUS ruling illustrates, the practical reality that AI increasingly produces the substantive inventive content forces patent systems to resort to preservation through substantive thinning and attribute inventorship to humans even though their cognitive contribution is attenuated or merely nominal.

The immediate problems with this pragmatic fudge are that it rewards obfuscation, penalises candour, and produces a patent landscape in which formal attribution bears an increasingly tenuous relationship to actual processes of innovation. Yet the deeper problem is that it entrains a legitimation deficit. If the named inventor’s contribution consists merely in recognising an AI system’s output, both the ownership and the desert inferences lose their grounding: the invention is not the product of the named person’s mind in any robust sense, and their creative labour therefore cannot be what generates a moral claim to recognition or compensation.

This legitimacy deficit is likely to be exacerbated by the distributive consequences of AI-generated patenting. The capacity to generate patentable output at scale will tend to be concentrated in a small number of firms with the resources to run general-purpose discovery systems. A system that knowingly grants monopolies on the basis of nominal human inventorship alone – while the actual inventive work is done by AI systems owned by a concentrated group of firms – reintroduces just the rent-seeking problem that the Statute of Monopolies sought to remedy.⁷

5.4 Dispute Resolution

The dispute resolution function, too, is threatened if AI-generated inventions come to be regarded as patentable. Following the America Invents Act (AIA) of 2011, the United States abandoned its first-to-invent system in 2013 – but it adopted *first-inventor-to-file*, not first-to-file like the rest of the world. Under a pure first-to-file regime, the cognitive act of conception would lose its procedural relevance entirely. But the “inventor” in “first-inventor-to-file” does residual work: the date of conception continues to play a role in *derivation proceedings* under 35 U.S.C. § 135, which aim to prevent applicants from patenting ideas they obtained from the true inventor.

It is precisely derivation disputes that AI-generated inventions complicate. When a chemist trained in an established field conceives a molecule, the distinction between what they derived from others and what they contributed originally is challenging, but the evidentiary resources of laboratory records and documented communication between the alleged inventor and deriver make the inquiry tractable. With AI systems, however, this evidentiary framework threatens to lose its grip because derivation becomes too *diffuse*. When derivation passes through the latent space of a massive neural network, supported perhaps by Retrieval-Augmented Generation (RAG), the interpersonal transmission of a discrete mental state is

⁷ In philosophical terms, this threatens to systematically violate the Lockean proviso (Nozick 1974; Shiffrin 2001); see also Hettinger (1989) on the tension between intellectual property and public interest; on the problem that the concentration of power in the hands of a small number of AI companies, see Queloz (2025a) and Korinek & Vipra (2025).

replaced by diffuse computational dependencies. The debt to the prior state of the art may become too thinly distributed to be traceable with any degree of certainty. The ideas of datable acts of conception and communication, built for disputes between human individuals, struggle to get a grip on the forms of derivation enabled by AI.

6. Conceptual Adaptation through Principled Disaggregation

The inherited concept thus presents us with a dilemma. If we deny patents for AI-generated inventions, we undermine incentivisation and dissemination. If we grant patents with nominal human inventors, we undermine legitimisation, dispute resolution, and the deeper aims of dissemination. In either case, the concept of inventor fails to discharge the very functions that animate its use.

Yet the point is emphatically not that the inventor concept has become obsolete or that the needs it has historically met have evaporated. Quite the contrary: those needs are as pressing as ever. The argument is that the inherited concept is becoming less capable of serving those needs effectively.

Two responses dominate the current debate, and both are inadequate. The first – call it the *expansionist* response – proposes to extend inventorship to AI systems, either by reinterpreting existing statutes or through new legislation granting AI systems legal inventor status Abbott (Abbott 2020). But this imports anthropocentric assumptions about human understanding and intellectual creation – and the ownership and desert inferences that accompany them – into a domain where they do not obviously have a purchase.

The second response – the *preservationist* response – insists that existing law correctly requires human inventorship and that the appropriate response to AI-generated inventions is either to deny them patent protection or to locate a human inventor by generously interpreting human contribution (the Swiss approach). But preservationism results either in innovation-suppressing rigidity or in a legitimacy-undermining fudge.

What expansionism and preservationism share is more revealing than what divides them, however. Both accept that *inventor* must remain a unified concept – they disagree only about

its extension. But this shared premise is precisely what the genealogy calls into question. The concept's unity is not the product of principled design; it is the sediment of six centuries of accretion, in which distinct practical pressures – attracting productive capacity, filtering out rent-seeking, extracting tacit knowledge, honouring creative genius, resolving temporal priority – forged four distinct inferences and four distinct functions into a single concept. To ask whether AI falls inside or outside this concept is to presuppose that can otherwise remain as it is.

The pragmatic genealogy suggests a third path, opened up by asking whether the concept should remain unified at all. If the concept bundles together four inferences serving four distinct social functions, and if AI disrupts this inferential pattern's ability to discharge those functions, then the right response is not to expand or preserve this multifunctional bundle, but to *disaggregate* it: to divide the conceptual labour across distinct concepts, each tailored to a specific function.

While a Swiss Army knife has its uses, we prefer dedicated instruments for high-stakes craftsmanship. Similarly, a concept that has accrued a plurality of functions over centuries may need to be split into distinct concepts, each more attuned to a specific end than a single concept torn between multiple roles can be. J.B. Maund already advocated this strategy, which he dubbed *conceptual fission* (1981), for colour concepts.⁸ But it is not unprecedented in legal and political thought either. Krasner (1999) has argued that “sovereignty” bundles together at least four functionally distinct attributes – domestic authority, interdependence control, international legal recognition, and non-intervention – that can and routinely do come apart;⁹ and Cueni (2020) has advocated a functional disaggregation strategy for legitimating international institutions. The concept of “property” itself came to be understood in terms of a

⁸ Scharp (2013) also advocates splitting the concept of truth into two specialised concepts.

⁹ Walker (2003) has suggested that the history of European constitutionalism is in part a history of functionally differentiating these attributes across competing and overlapping sites of authority; Grimm (2009) gives a genealogical account of how the concept of sovereignty accumulated multiple functions through historically contingent pressures. The phrase “disaggregation” is associated with Slaughter (2004).

“bundle of sticks” model that separates use rights, exclusion rights, transfer rights, and income rights.¹⁰

6.1 Incentivisation Without Creative Origination

The disruption diagnosed in § 5.1 pointed to a fundamental misalignment: the inherited concept channels incentives through the act of individual conception, but the activities that most need incentivising in the case of AI-driven innovation lie upstream and downstream of this. What is needed is a way to direct incentives towards the actual locus of investment risk without having to route them through the fiction of an individual human act of conception.

Once we recognise that the incentivisation function need not be served by the same concept as the legitimisation, dissemination, and dispute resolution functions, we can ask directly: what mechanisms would most effectively incentivise the investments that drive AI-powered innovation? The genealogy provides a principled justification for this: the concept of inventor was originally constructed to incentivise the introduction of new technical knowledge, and this concern is served equally well by directing incentives to whoever performs the innovation-enabling activities as by the inherited concept’s insistence that reward follow individual creative origination. After all, the genealogy suggests that the Enlightenment-Romantic individualisation of inventorship and its subsequent equation with an individual act of creative origination was responsive to needs for legitimisation and dispute resolution, and had nothing to do with incentivisation.

Several possibilities merit consideration. The first is *structured exclusivity*. Rather than granting the same twenty-year monopoly regardless of how an invention was produced, patent terms could be calibrated to reflect the actual investment required. AI-generated inventions produced at low marginal cost through general-purpose discovery methods might receive shorter terms or narrower scope, while inventions requiring substantial downstream investment might receive the traditional term or longer. The guiding principle would be

¹⁰ See Munzer (1990) for an influential and philosophically rigorous statement of a bundle theory of property drawing on Hohfeld (1913).

proportionality (Merges, 2011): the strength of the incentive should track the magnitude of the investment it is designed to elicit.

Second, *compulsory licensing provisions* allow patent holders to retain the patent, but require them to license the invention to others at a regulated, reasonable royalty. Such provisions already exist in many jurisdictions for pharmaceuticals, and are triggered by conditions such as public health emergencies. For AI-generated inventions produced at low marginal cost, compulsory licensing could be triggered more broadly: if the invention was generated by general-purpose AI methods and required minimal downstream investment, such provisions could ensure that competitors can obtain licenses, rewarding the patent holder without the public bearing the full cost of a twenty-year monopoly on an invention whose marginal cost was trivial.

A third possibility is *direct incentivisation* of infrastructure investment: if the bottleneck to AI-enabled innovation is upstream – in compute, data, and verification – then upstream investments could be incentivised directly rather than indirectly through patents on outputs. Mechanisms might include process patents on discovery pipelines, investment tax credits, or data-sharing incentives that reward firms for contributing to publicly accessible training datasets.

6.2 Dissemination Without Understanding

As the genealogy showed, the historical shift to written specifications forged a tight link between inventorship and communicable understanding. Because human innovators typically generate solutions by understanding them in a principled way, the act of discovery naturally equips them to explain their inventions in writing. With AI models, however, this convergence between competence and communicability breaks down. The opaque internal mechanics driving their discoveries resist translation into the linear, principled format of a patent specification (Sullivan 2022; Beisbart 2025; Queloiz and Beckmann 2026).

A disaggregated framework resolves this tension by decoupling discovery from disclosure. If the dissemination function is to be served, the system must separate the requirement to

articulate the invention itself – what we might call *substantive disclosure* – from the requirement to illuminate the process that generated it – what we might call *process disclosure*.

First and foremost, the framework must secure substantive disclosure. Regardless of how the invention was produced, the applicant must provide a description of the invention itself enabling a “person having ordinary skill in the art” to reproduce and build upon it. Because the AI cannot necessarily supply such an explanation, however, the burden falls on the human deployers to reverse-engineer the AI’s output until it becomes communicable technical knowledge. This represents a modernised “duty to teach”. It may require extensive downstream investment: physical prototyping, clinical validation, or the use of secondary “explainer” AI models explicitly trained to reverse-engineer human-interpretable physical principles from the primary model’s outputs. If it is to serve the dissemination function, however, the patent cannot be granted until the invention itself is sufficiently well understood.

While the invention itself is the primary target of the patent bargain, the AI discovery infrastructure may itself represent a technology of immense public interest. To prevent firms from monopolising the technology that generates inventions, a disaggregated framework can demand supplementary disclosures about the generative process. We can distinguish three aspects of process disclosure:

- a) *Contextual*: what the human operators understood about the problem space, the constraints they imposed, and the reward functions they designed to direct the AI.
- b) *Architectural*: specification of the AI system’s architecture, training data, input parameters, hyperparameters, and selection criteria – everything needed for a skilled practitioner to reproduce the inventive technology itself.
- c) *Interpretational*: whatever mechanistic understanding the human deployers possess of how and why the system produced the output, drawing on the evolving resources of mechanistic interpretability and other explainable AI methods.

This two-tiered approach adapts the historic duty to teach to an era where the entity doing the discovering is no longer necessarily the entity capable of doing the explaining, while ensuring that the broader methodological advances of AI-driven research are themselves disseminated.

6.3 Legitimation Beyond Ownership and Desert

The disruption diagnosed in § 5.3 showed that nominal human inventorship deprives the ownership and desert inferences of their grounding, leaving monopolies unjustified.

The genealogy also reveals, however, that ownership and desert are not the only available bases for legitimation; historically speaking, they are relatively recent additions. The Venetian layer legitimated monopolies instrumentally: they rewarded the *introduction* of productive capacity, regardless of who cognitively originated it. The English layer legitimated them as a *quid pro quo*: exclusivity in exchange for the cost and effort of bringing an industry to the realm and teaching it to others. These older, more consequentialist justifications survive the shift to AI-generated invention.

But a disaggregated legitimation function cannot simply default to a pure consequentialism either. The view that monopolies are justified whenever granting them maximises innovation provides no principled constraint on the scope or duration of the monopoly – it would justify perpetual monopolies if the incentive calculus came out right. The inherited concept's desert-based legitimation, whatever its other difficulties, provided such a constraint: the monopoly was proportionate to the creative labour invested. What is needed in the AI context is an analogous constraint grounded in different presuppositions.

Between the extremes of pure consequentialism and Romantic creative desert lies a robust middle ground: *contribution-based legitimation*. Rather than justifying patents by appealing to individual creativity, we can justify them based on how much identifiable human effort made a difference. This is a counterfactual standard: it asks what would have been produced without this human contribution. If the product of the human-AI collaboration measurably exceeds this baseline, then granting a patent can be legitimate. Under this framework, the monopoly is still granted to human or corporate deployers, who are responsive to such incentives, and it is

calibrated to the degree and type of their material and directional contribution – regardless of whether that contribution is “creative” in the traditional cognitive sense.

Thus, legitimation would shift from romantic narratives to a more accountable, socially grounded measure of human input. The human or firm who invested in the compute infrastructure, curated bespoke training datasets, mathematically formulated a novel problem domain, and rigorously verified the AI system’s output has made a genuine, indispensable contribution to bringing the innovation to the public. They have acted much like the Venetian importers of knowledge: they bore a risk and expended resources to introduce a capability the public lacked. Disaggregation allows us to formally sever the *entitlement to the patent* from the *status of cognitive inventor*. We can legitimate a human’s monopoly on the basis of their infrastructural and curational contribution, without having to pretend that they experienced a “flash of genius”.

The contribution-based standard also has the virtue of being sensitive to the problem of low marginal discovery costs, by presenting a monopoly as legitimate only if it is proportionate to what the applicant added beyond what the generally available discovery infrastructure would have produced in short order anyway.

6.4 Dispute Resolution Beyond the “First and True Inventor”

The disruption diagnosed in § 5.4 showed that AI fundamentally undermines the evidentiary framework of inherited dispute resolution. Traditional priority and derivation proceedings rely on reconstructing timelines of human intellectual activity (“When was the invention conceived?”) and communications (“Who told what to whom?”). When inventive steps are distributed across massive training corpora, latent spaces, and real-time algorithmic retrieval, this evidentiary framework loses its grip.

The disaggregated response addresses these difficulties by separating two issues that the inherited concept conflated into the single figure of the “first and true inventor”: *temporal priority* on the one hand and *causal provenance* on the other. By decoupling these, dispute

resolution can be reconceived from the ground up around the realities of computational discovery.

Retaining pure “first-to-file” and “first-to-conceive” criteria invites the “IP land-grab” warned against in § 5.1. If we disaggregate, however, temporal priority no longer needs to be anchored in the initial cognitive generation of an idea. Since the activities that now most require incentivising are upstream infrastructure investments and downstream validation, priority rules should be recalibrated to track *substantiation* rather than mere generation. In a race between two firms using AI to discover a new semiconductor material, the dispute should not be resolved by whose server output the chemical formula a millisecond earlier, but by who first satisfied the modernized “duty to teach” (§ 6.2): who first physically prototyped it, verified its properties, and submitted a replicable specification. Shifting the priority standard to “first-to-substantiate” neutralises the threat of automated patent spamming, rendering high-speed AI generation legally inert until combined with tangible investment into its substantiation.

The second aspect of dispute resolution – the derivation concern – requires an even more radical conceptual shift. Inquiries into human cognitive derivation must be replaced by data-forensic standards of provenance. This requires transposing legal concepts of derivation into the technical language of machine learning. For instance, a model “memorises” when it reconstructs its training data verbatim, but “generalises” when it infers underlying principles to generate novel solutions. This technical distinction can provide a modern analogue for the legal boundary between derivation and independent invention.

Similarly, while retrieval-augmented generation (RAG) conceptually blurs the line between derivation and invention, it actually makes the evidentiary problem more tractable. Unlike the human mind, a RAG system leaves an exact log of which external knowledge bases were accessed and when. These RAG logs could replace traditional lab notebooks in derivation proceedings.

A disaggregated framework can thus address priority disputes by requiring transparency about the inputs to AI-generated invention and by developing appropriation standards calibrated to these new forms of intellectual dependency.

7. Conclusion: Reinventing the Inventor

This paper has advanced a principled case for disaggregating the concept of “inventor”. As we have inherited it, the concept bundles together four inferential roles – attributing ownership, desert, entitlement to a patent, and responsibility to disclose – which serve four social functions: incentivising innovation, legitimating monopolies, disseminating knowledge, and resolving priority disputes. These roles and functions converged on a single human individual only because, historically, all four pointed to the same person. The advent of generative AI has broken this convergence.

The genealogy has revealed this predicament with distinctive clarity by reconstructing the sequence of practical needs that shaped the concept’s formation: the Venetian coordination problem through the English duty to teach, the Statute of Monopolies’ anti-rent-seeking filter, the written specification’s forging of a link between inventorship and understanding, the Romantic layer’s addition of ownership and creative desert as legitimacy devices, and the corporate era’s strategy of preservation through substantive thinning. This showed that the concept accumulated its cognitive presuppositions and functions for historically intelligible reasons. But it also brought out that many of those reasons have now lapsed: knowledge is no longer embodied primarily in persons; rent-seeking is now curtailed by other mechanisms; the first-to-invent system was abandoned in 2013; and the Enlightenment-Romantic inferences to ownership and desert are precisely what the advent of knowledge-creating AI systems most directly challenges.

The needs that justify having an inventor concept – to incentivise innovation, legitimate monopolies, disseminate knowledge, and resolve disputes – remain as pressing as ever. But our capacities and circumstances have changed so profoundly that the inherited concept is now an unreliable instrument for meeting those needs.

The response I have advocated is not to extend the inherited concept to AI systems, which would import Romantic assumptions about ownership and creative desert into a domain where they lack purchase. Nor can we afford to preserve it unchanged, which produces either

innovation-suppressing rigidity or a legitimacy-undermining fiction. Instead, we must disaggregate the concept into its functional components and develop separate conceptual resources for incentivisation, dissemination, legitimation, and dispute resolution.

A natural worry about disaggregation is that it replaces a workable – if imperfect – concept with a proliferation of new concepts that are harder to administer, coordinate, and understand. Patent law values administrability, and having “inventor” as a single category enables procedural economy. But the genealogy undercuts this objection. The bundled concept was not designed for administrability; it *accrued* its multiple functions through a series of practical pressures. Moreover, the costs of continuing to address these multiple functions with a single concept – costs in obfuscation, in legitimacy deficits, and in innovation forgone – may soon outweigh the administrative friction of disaggregation.

In fact, disaggregation is already happening, if only piecemeal and without principled guidance. The shift away from first-to-invent has already loosened the link between dispute resolution and inventorship’s cognitive presuppositions. Employee invention laws, which vest patent ownership in the employer while naming the employee as inventor, have already disconnected ownership from inventorship. The USPTO’s 2024 guidance on AI-assisted inventions (USPTO, 2024) was a form of implicit disaggregation: while formally preserving the constraint that only a natural person could be named the inventor, it applied the *Pannu* factors to AI-assisted inventions, meaning that a human only had to show that they made a “significant inventive contribution” to the AI’s output to be an inventor; this was a step away from the inherited concept’s assumption that a single inventor conceived the whole (though the USPTO has since rescinded this and reaffirmed the traditional “conception” test in its November 2025 “Revised Inventorship Guidance for AI-Assisted Inventions”). And the Swiss court’s acceptance that Thaler qualified as inventor on the grounds of “sufficient overall influence” over the AI system is an implicit recognition of the need for disaggregation: the court formally preserved the requirement of an “intellectual act of creation”, yet the involvement it actually accepted – providing data, training the system, and recognising the output as patentable – amounts to just the kind of *contribution-based standard* I proposed rather than the creative

origination the doctrine nominally demands. The legitimation function is already being discharged on new grounds, even as the old vocabulary is retained.

What these developments lack is a principled framework that unifies them into a coherent strategy. The pragmatic genealogy of inventorship I have offered here can act as a guide towards such a strategy and pave the way for systematic and self-conscious disaggregation.

Viewed from inside the inherited concept, the question “Can AI be an inventor?” looks like a profound metaphysical inquiry about the nature of invention. Viewed genealogically, however, it is a practical question about whether a concept forged to break Renaissance guild monopolies and curb seventeenth-century royal prerogatives still serves us well in the age of generative AI. In building artificial neural networks that can make discoveries, we have effectively invented an inventor. We now need to remember that the very notion of the inventor is itself something we invented. The time has come to reinvent it.

Bibliography

- Abbott, Ryan. 2020. *The Reasonable Robot: Artificial Intelligence and the Law*. Cambridge: Cambridge University Press.
- Abbott, Ryan B. 2017. ‘Patenting the output of autonomously inventive machines’. *Landslide* 10: 16.
- Baudry, Jérôme. 2020. ‘A Politics of Intellectual Property: Creating a Patent System in Revolutionary France’. *Technol Cult* 61 (4): 1017-1044.
- Baudry, Marc, and Béatrice Dumont. 2017. *Patents: prompting or restricting innovation?* Vol. 12. Hoboken, NJ: John Wiley & Sons.
- Beckmann, Pierre, and Matthieu Queloz. 2025. ‘Mechanistic Indicators of Understanding in Large Language Models’. *arXiv* arXiv:2507.08017: 1–32.
- Beisbart, Claus. 2025. ‘Digging deeper with deep learning? Explanatory understanding and deep neural networks’. *European Journal for Philosophy of Science* 15 (3): 40.
- Boldrin, Michele, and David K Levine. 2013. ‘The case against patents’. *Journal of Economic Perspectives* 27 (1): 3-22.
- Bracha, Oren. 2016. *Owning Ideas: The Intellectual Origins of American Intellectual Property, 1790–1836*. Cambridge: Cambridge University Press.
- Bundesgerichtshof. 2024. President of the German Patent and Trademark Office v. Stephen L. Thaler (DABUS). Karlsruhe: Bundesgerichtshof (German Federal Court of Justice).
- Cocquyt, T., Z. Zhou, J. Plomp, and L. van Eijck. 2021. ‘Neutron tomography of Van Leeuwenhoek’s microscopes’. *Science Advances* 7 (20).
- Court of Common, Pleas. 1778. *Liardet v. Johnson*. London: Court of Common Pleas.
- Cueni, Damian. 2020. *Public Law Analogies in International Legal Theory*. Doctoral Thesis, Rechtswissenschaftliches Institut, University of Zurich.
- Cunningham, Tom. 2026. “Knowledge-Creating LLMs.” *tecunningham.github.io*, 2026. tecunningham.github.io/posts/2026-01-29-knowledge-creating-llms.html.

- Doron, Guy, Sam Genway, Mark Roberts, and Sai Jasti. 2025. 'Generative AI: driving productivity and scientific breakthroughs in pharmaceutical R&D'. *Drug Discovery Today* 30 (1): 104272.
- European Patent Office. 2020. Grounds for the EPO's Decision to Refuse European Patent Application 18 275 163.6 (DABUS). Munich: European Patent Office.
- Frumkin, Maximilian. 1945. 'The origins of patents'. *Journal of the Patent Office Society* 27 (3): 143-49.
- Full Court of the Federal Court of Australia. 2022. *Commissioner of Patents v. Thaler*. Canberra: Full Court of the Federal Court of Australia.
- Grimm, Dieter. 2009. *Souveränität : Herkunft und Zukunft eines Schlüsselbegriffs*. Berlin: Berlin University Press.
- Hegel, Georg Wilhelm Friedrich. 1970. *Grundlinien der Philosophie des Rechts*. Vol. 7, *Theorie-Werkausgabe*. Frankfurt am Main: Suhrkamp.
- Heon, Lexi. 2022. 'Artificially obvious but genuinely new: how artificial intelligence alters the patent obviousness analysis'. *Seton Hall L. Rev.* 53: 359.
- Hettinger, Edwin C. 1989. 'Justifying intellectual property'. *Philosophy & Public Affairs*: 31-52.
- Hilaire-Pérez, Liliane. 2010. *L'invention technique au siècle des Lumières*. Paris: Albin Michel.
- Hobbes, Thomas. 2006. *Leviathan or The Matter, Forme and Power of a Common Wealth Ecclesiasticall and Civil*. Edited by G. A. J. Rogers and Karl Schuhmann. London: Continuum.
- Hohfeld, Wesley Newcomb. 1913. 'Some Fundamental Legal Conceptions as Applied in Judicial Reasoning'. *The Yale Law Journal* 23 (1): 16-59.
- Hulme, E Wyndham. 1897. 'On the Consideration of the Patent Grant Past and Present'. *Law Quarterly Review* 13: 313.
- Hulme, E. Wyndham. 1896. 'History of the Patent System Under the Prerogative and at Common Law'. *Law Quarterly Review* 12 (2): 141-154.
- Hume, David. 2000. *A Treatise of Human Nature*. Edited by David Fate Norton and Mary J. Norton. Oxford: Oxford University Press.

- Isomorphic Labs Team. 2026. 'Accurate Predictions of Novel Biomolecular Interactions with IsoDDE'.
- Khan, B. Zorina. 2019. *The Democratization of Invention: Patents and Copyrights in American Economic Development, 1790–1920*. Cambridge: Cambridge University Press.
- Korinek, Anton, and Jai Vipra. 2025. 'Concentrating intelligence: scaling and market structure in artificial intelligence'. *Economic Policy* 40 (121): 225-256.
- Kostylo, Joanna. 2010. 'From Gunpowder to Print: The Common Origins of Copyright and Patent'. In *Privilege and Property*, 21–50. Cambridge: Open Book Publishers.
- Krasner, Stephen D. 1999. *Sovereignty: Organized Hypocrisy*. Princeton: Princeton University Press.
- Lim, Daryl. 2018. 'AI & IP: innovation & creativity in an age of accelerated change'. *Akron L. Rev.* 52: 813.
- Locke, John. 2003. *Two Treatises of Government and A Letter Concerning Toleration*. Edited by Ian Shapiro. New Haven: Yale University Press.
- MacLeod, Christine. 1988. *Inventing the Industrial Revolution: The English Patent System, 1660–1800*. Cambridge: Cambridge University Press.
- MacLeod, Christine. 2007. *Heroes of Invention: Technology, Liberalism and British Identity, 1750–1914*. Cambridge: Cambridge University Press.
- Madika, Benediktus, Aditi Saha, Chaeyul Kang, Batzorig Buyantogtokh, Joshua Agar, Chris M Wolverton, Peter Voorhees, Peter Littlewood, Sergei Kalinin, and Seungbum Hong. 2025. 'Artificial intelligence for materials discovery, development, and optimization'. *ACS nano* 19 (30): 27116-27158.
- Mandich, Giulio. 1948. 'Venetian Patents (1450-1550)'. *Journal of the Patent Office Society* 30 (3): 166–224.
- Maund, J. B. 1981. 'Colour: A Case for Conceptual Fission'. *Australasian Journal of Philosophy* 59 (3): 308–322.

- Merchant, Amil, Simon Batzner, Samuel S. Schoenholz, Muratahan Aykol, Gowoon Cheon, and Ekin Dogus Cubuk. 2023. 'Scaling deep learning for materials discovery'. *Nature* 624 (7990): 80-85.
- Merges, Robert P. 2011. *Justifying Intellectual Property*. Cambridge, MA: Harvard University Press.
- Merritt, Cole G. 2023. 'A compulsory solution to the machine problem: Recognizing artificial intelligence as inventors in patent law'. *Vand. J. Ent. & Tech. L.* 25: 211.
- Molà, Luca. 2000. *The Silk Industry of Renaissance Venice*. Baltimore, London: Johns Hopkins University Press.
- Moore, Adam D. 2002. 'Intellectual property, innovation, and social progress: the case against incentive based arguments'. *Hamline L. Rev.* 26: 601.
- Moser, Petra. 2013. 'Patents and Innovation: Evidence from Economic History'. *The Journal of Economic Perspectives* 27 (1): 23-44.
- Mossoff, Adam. 2012. 'Saving Locke from Marx: The Labor Theory of Value in Intellectual Property Theory'. *Social Philosophy and Policy* 29 (2): 283-317.
- Munzer, Stephen R. 1990. *A Theory of Property*. Cambridge: Cambridge University Press.
- Nemec, Douglas R., and Laura M. Rann. 2023. "AI and Patent Law: Balancing Innovation and Inventorship." *Skadden Insights*.
- Novikov, Alexander, Ngàn Vũ, Marvin Eisenberger, Emilien Dupont, Po-Sen Huang, Adam Zsolt Wagner, Sergey Shirobokov, Borislav Kozlovskii, Francisco JR Ruiz, and Abbas Mehrabian. 2025. 'AlphaEvolve: A coding agent for scientific and algorithmic discovery'. *arXiv preprint arXiv:2506.13131*.
- Nozick, Robert. 1974. *Anarchy, State, and Utopia*. Oxford: Blackwell.
- Pottage, Alain, and Brad Sherman. 2010. *Figures of Invention: A History of Modern Patent Law*. Oxford: Oxford University Press.
- Queloz, Matthieu. 2021. *The Practical Origins of Ideas: Genealogy as Conceptual Reverse-Engineering*. Oxford: Oxford University Press.

- Queloz, Matthieu. 2025a. 'Dropping Anchor in Rough Seas: Co-Reasoning with Personalized AI Advisors and the Liberalism of Fear'. *Philosophy & Technology* 38 (4): 170.
- Queloz, Matthieu. 2025b. *The Ethics of Conceptualization: Tailoring Thought and Language to Need*. Oxford: Oxford University Press.
- Queloz, Matthieu. 2026. 'Needs of the Mind: How Aptic Normativity Can Guide Conceptual Adaptation'. *Philosophical Studies*.
- Queloz, Matthieu, and Pierre Beckmann. 2026. 'Why We Care About Understanding: Competence through Predictive Compression'. *manuscript*: 1–33.
- Reinbold, Patric M. 2020. 'Taking artificial intelligence beyond the turing test'. *Wis. L. Rev.*: 873.
- Romm, Connor. 2021. 'Putting the Person in Phosita: The Human's Obvious Role in the Artificial Intelligence Era'. *BCL Rev.* 62: 1413.
- Scharp, Kevin. 2013. *Replacing Truth*. Oxford: Oxford University Press.
- Sherman, Brad, and Lionel Bently. 1999. *The Making of Modern Intellectual Property Law*. Vol. 1. Cambridge: Cambridge University Press.
- Shiffrin, Seana Valentine. 2001. 'Lockean Theories of Intellectual Property'. In *New Essays in the Political Theory of Property*. Edited by Stephen R. Munzer, 138–167. Cambridge: Cambridge University Press.
- Slaughter, Anne-Marie. 2004. *A New World Order*. Princeton: Princeton University Press.
- Sullivan, Emily. 2022. 'Understanding from Machine Learning Models'. *The British Journal for the Philosophy of Science* 73 (1): 109–133.
- Swiss Federal Administrative Court. 2025. Stephen L. Thaler v. Swiss Federal Institute of Intellectual Property. Bern: Swiss Federal Administrative Court (Bundesverwaltungsgericht).
- Tobias, Alexander V, and Adam Wahab. 2025. 'Autonomous 'self-driving' laboratories: a review of technology and policy implications'. *Royal Society Open Science* 12 (7): 250646.
- U. K. Supreme Court. 2023. Thaler v. Comptroller-General of Patents, Designs and Trade Marks. London: UK Supreme Court.

- United States Court of Appeals for the Federal, Circuit. 2022. *Thaler v. Vidal*. Washington, D.C.: United States Court of Appeals for the Federal Circuit.
- United States Court of, Customs, and Appeals Patent. 1929. *Townsend v. Smith*. Washington, D.C.: United States Court of Customs and Patent Appeals.
- United States Supreme, Court. 1941. *Cuno Engineering Corp. v. Automatic Devices Corp.* Washington, D.C.: United States Supreme Court.
- Usselman, Steven W. 2013. “Research and Development in the United States Since 1900: An Interpretive History.” Economic History Workshop.
- Waldron, Jeremy. 2013. “To Bestow Stability upon Possession’: Hume’s Alternative to Locke’. In *Philosophical Foundations of Property Law*. Edited by James Penner and Henry Smith, 1–12. Oxford University Press.
- Walker, Neil. 2003. ‘Late Sovereignty in the European Union’. In *Sovereignty in Transition*. Edited by Neil Walker, 3–32. Oxford: Hart.
- Walterscheid, Edward C. 1997. ‘To Promote the Progress of Useful Arts: American Patent Law and Administration, 1787-1836 (Part I)’. *Journal of the Patent and Trademark Office Society* 79 (1): 61–79.
- Wang, Hanchen, Tianfan Fu, Yuanqi Du, Wenhao Gao, Kexin Huang, Ziming Liu, Payal Chandak, Shengchao Liu, Peter Van Katwyk, and Andreea Deac. 2023. ‘Scientific discovery in the age of artificial intelligence’. *Nature* 620 (7972): 47-60.
- Williams, Bernard. 2005. *In the Beginning Was the Deed: Realism and Moralism in Political Argument*. Edited by Geoffrey Hawthorne. Princeton: Princeton University Press.
- Woodbury, Levi, and District of Massachusetts Circuit Court. 1845. *Davoll v. Brown*. Boston: Circuit Court, District of Massachusetts.
- Woodmansee, Martha. 1984. ‘The Genius and the Copyright: Economic and Legal Conditions of the Emergence of the ‘Author’’. *Eighteenth-Century studies* 17 (4): 425–448.
- Woodmansee, Martha. 1992. ‘On the author effect: Recovering collectivity’. *Case Legal Studies Research Paper* (1992): 10.

Yuksekgonul, Mert, Daniel Kocaja, Xinhao Li, Federico Bianchi, Jed McCaleb, Xiaolong Wang, Jan Kautz, Yejin Choi, James Zou, and Carlos Guestrin. 2026. ‘Learning to discover at test time’. *arXiv preprint arXiv:2601.16175*.